Project Cinema — Product Requirements Document

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1. Vision

A personalised movie recommendation system where users rate films, write reviews, and receive intelligent recommendations that improve over time. An LLM layer handles natural language input ("I want something like Parasite but funnier") and explains every recommendation in plain English. Built as a portfolio project first, designed to scale into a real product.

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2. Problem Statement

Existing recommenders (Netflix, Letterboxd) are black boxes — they tell you what to watch but never why. Users have no way to express nuanced preferences in natural language, and recommendations don't visibly improve as the user engages more. This project solves both: transparent, explainable recommendations driven by real user behaviour and conversational input.

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3. Target Users

User Type	Description
New user	No history. Onboarded via genre/mood preferences. Gets content-based recs immediately.
Returning user	Has rated 5+ movies. Gets collaborative filtering recs that improve over time.
Power user	Has rated 20+ movies and written reviews. Gets hybrid recs + full LLM explanations.

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4. Core Features

4.1 Rating System — 5-Tier Visual Scale

Each tier has a distinct visual identity — icon, colour, and label. No stars, no numbers.

Tier	Icon	Colour	Meaning
Peak Cinema	🔥 Flame	Deep amber `#C8821A`	Absolute best — life-changing
Masterpiece	✦ Diamond	Rich gold `#D4A017`	Exceptional craft
Great Watch	◎ Circle	Warm teal `#2D9CDB`	Solid, enjoyable
Mid	— Dash	Muted sand `#8A7560`	Forgettable
Skip	× Cross	Faded rust `#8B4513`	Not worth your time

Visual treatment:

- Tiers displayed as pill badges with icon + label
- On rating, selected tier animates with a brief pulse/glow
- On movie cards, tier badge sits top-right with subtle background fill
- Tier history shown as a visual distribution bar on user profile

4.2 Genre Visual System

Each genre has a unique icon and colour tag — never plain text labels.

Genre	Icon	Colour
Action	⚡	Orange `#E8712A`
Drama	🎭	Deep blue `#1A5A8A`
Thriller	👁	Dark crimson `#8B0000`
Comedy	☀	Bright yellow `#F5C518`
Sci-Fi	◈	Electric teal `#00B4D8`
Horror	☽	Deep violet `#4A0E8F`
Romance	♡	Rose `#E8436A`
Animation	✦	Lime `#7CB518`
Documentary	◉	Slate `#4A6FA5`
Crime	⊙	Charcoal `#2C2C2C`
War	✦	Army green `#4A5240`
Musical	♪	Magenta `#C7369A`

Displayed as compact icon+label chips on movie cards. Clickable to filter.

4.3 Recommendation Engine (ML Core)

Three-layer hybrid pipeline:

Layer	Model	Purpose
Content-based	Random Forest	New users, cold-start. Uses movie metadata + genre features.
Collaborative filtering	SVD (Matrix Factorization)	Warm users. Learns from rating patterns across all users.
Hybrid fusion	Weighted ensemble	Blends CF + content scores. CF weight increases as user rates more.

4.4 LLM Layer (Claude API)

Two roles:

Input handler — User types natural language preferences:

> "I'm in the mood for something dark and psychological, not too long"

LLM extracts structured intent (genres, mood, runtime, tone) and passes it to the recommendation engine as a feature query.

Output explainer — For every recommendation, LLM generates a 2–3 sentence explanation:

> "We think you'll love this because you rated Parasite Peak Cinema — both share a slow-burn thriller structure with dark social commentary. 94% of users with your taste profile rated it Masterpiece or above."

4.5 Explainability — Top 3 Recommendation Cards

Instead of a bar chart, the XAI output is presented as **3 cinematic recommendation cards** displayed side by side (or stacked on mobile). Each card contains:

- Movie poster placeholder (or TMDB poster via links.csv)
- Title + year + genre chips
- Predicted tier badge (with icon + colour)
- A single bold "match reason" headline — e.g. *"Because you loved Parasite"*
- 2–3 sentence LLM explanation below the headline
- A subtle confidence indicator — e.g. a glowing arc or fill bar unique to each card
- "Add to watchlist" + "Rate it" actions

Card design language:

- Warm parchment background (Dune palette — see Section 7)
- Cards have a faint warm glow on hover
- The top pick (rank 1) is slightly larger / more prominent than 2 and 3
- Revealed with a staggered fade-in animation

4.6 Review System

- Short-form text reviews (optional, attached to a rating)
- Displayed on movie pages with tier badge + username
- Reviews feed into sentiment analysis in Phase 3

4.7 User Profile

- Taste fingerprint: top genres (shown as icon chips), favourite tier distribution
- Rating history with tier badges
- Recommendation feed that updates as user rates more

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5. Design System — Dune Palette (Light Mode)

Inspired by the warm desert tones, spice gold, and Fremen blue of Dune.

Token	Value	Usage
`--bg`	`#E8D5A3`	Page background
`--surface`	`#F2E4C0`	Cards, panels
`--surface-2`	`#EAD4A0`	Hover states
`--border`	`#C8A85A`	Borders
`--accent`	`#B07010`	Primary actions
`--accent-deep`	`#8A5208`	Headings, strong text
`--spice`	`#C8821A`	Peak Cinema, highlights
`--fremen`	`#1A5A8A`	Links, info, Sci-Fi genre
`--fremen-bright`	`#2D9CDB`	Hover links, Great Watch tier
`--text`	`#1A1208`	Body text
`--muted`	`#6B4E18`	Secondary text
`--sand`	`#F5E8C0`	Input backgrounds

Dark mode: planned for Phase 3.

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6. Data Architecture

6.1 Training Data (Phase 1)

- **Source:** MovieLens ml-latest-small (100k ratings, 9k movies, 610 users)
- **Files:** ratings.csv, movies.csv, tags.csv, links.csv

6.2 User Data (Phase 2)

- New ratings appended to SQLite database
- Model retrains incrementally as new ratings come in
- Reviews stored as text, used for sentiment features in Phase 3

6.3 Feature Matrix

Feature	Source	Type
user_avg_rating	User history	Numeric
user_rating_count	User history	Numeric
user_rating_std	User history	Numeric
avg_rating	MovieLens aggregate	Numeric
rating_count	MovieLens aggregate	Numeric
release_year	movies.csv	Numeric
genre flags (8)	movies.csv	Binary
all_tags	tags.csv aggregated	Text (Phase 2)
sentiment_score	User reviews	Numeric (Phase 3)

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7. Technical Stack

Layer	Technology
ML models	scikit-learn (LR, RF), scipy (SVD)
LLM	Anthropic Claude API (claude-sonnet-4-6)
Explainability	SHAP (TreeExplainer)
Backend	Python + FastAPI
Database	SQLite (Phase 1) → PostgreSQL (Phase 2)
Frontend	React + Tailwind CSS
Deployment	Local (Phase 1) → Render / Railway (Phase 2)

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8. GitHub Repository Structure

```
project-cinema/
│
├── README.md                   # Project overview, setup, demo GIF
├── requirements.txt            # Python dependencies
├── .env.example                # API key template
│
├── data/
│   └── ml-latest-small/        # Raw MovieLens files (gitignored if large)
│       ├── ratings.csv
│       ├── movies.csv
│       ├── tags.csv
│       └── links.csv
│
├── notebooks/
│   └── cinema_recommender.ipynb   # Full training notebook
│
├── ml/
│   ├── train.py                # Model training pipeline
│   ├── predict.py              # Inference + top-N recommendations
│   ├── features.py             # Feature engineering
│   ├── evaluate.py             # Metrics + SHAP
│   └── models/                 # Saved model files (.pkl)
│       ├── rf_model.pkl
│       ├── lr_model.pkl
│       └── svd_matrix.pkl
│
├── api/
│   ├── main.py                 # FastAPI app
│   ├── routes/
│   │   ├── recommendations.py
│   │   ├── ratings.py
│   │   └── llm.py
│   └── db/
│       ├── database.py
│       └── schema.sql
│
├── frontend/
│   ├── public/
│   ├── src/
│   │   ├── components/
│   │   │   ├── TierBadge.jsx       # Peak Cinema · Masterpiece etc.
│   │   │   ├── GenreChip.jsx       # Icon + colour genre tags
│   │   │   ├── RecommendationCard.jsx  # Top 3 cards
│   │   │   ├── MovieCard.jsx
```

```
│   │   │   └── ReviewBox.jsx
│   │   ├── pages/
│   │   │   ├── Home.jsx
│   │   │   ├── Movie.jsx
│   │   │   └── Profile.jsx
│   │   ├── styles/
│   │   │   └── dune-theme.css      # Design tokens
│   │   └── App.jsx
│   └── package.json
│
└── docs/
    ├── PRD.md                  # This document
    └── architecture.png        # System diagram
```

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9. Phased Delivery Plan

Phase 1 — ML Core + Portfolio Demo *(now)*

- [ ] Clean training pipeline on ml-latest-small
- [ ] LR + RF + SVD models trained and evaluated
- [ ] SHAP explainability working
- [ ] Top 3 recommendation cards with LLM explanations (Claude API)
- [ ] Genre chip visual system
- [ ] Tier badge visual system
- [ ] Interactive React demo embedded in portfolio website
- [ ] Notebook published in `/notebooks`
- [ ] GitHub repo structured as above

Phase 2 — Full Web App

- [ ] FastAPI backend with rating + recommendation endpoints
- [ ] User accounts (simple auth)
- [ ] Ratings + reviews stored in SQLite
- [ ] React frontend with full Dune design system
- [ ] Model retraining triggered on new ratings
- [ ] Deployed on Render / Railway

Phase 3 — Real Product

- [ ] Sentiment analysis on reviews as ML feature
- [ ] Dark mode
- [ ] Social layer — see what friends rated
- [ ] PostgreSQL at scale

- [ ] TMDB poster integration via links.csv

Phase 4 — Mobile

- [ ] React Native app (iOS + Android)

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10. Success Metrics

Metric	Phase 1 Target	Phase 2 Target
RF classification accuracy	> 40% (5-class)	> 50% with user data
SVD RMSE	< 1.0	< 0.85
LLM explanation quality	Coherent + relevant	User rates useful > 70%
Cold-start coverage	100% new users get recs	—

> Note: 40–47% on a 5-class problem is strong — random baseline is 20%.

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11. What This Demonstrates (Resume Value)

Skill	Where it shows
End-to-end ML pipeline	Data → features → training → evaluation
Collaborative filtering	SVD matrix factorization
LLMs in production	Claude API for input parsing + output explanation
Explainable AI	SHAP values, top-3 card presentation
Real dataset	MovieLens (industry benchmark)
Product design	Design system, tier/genre visual language
Full-stack	FastAPI + React (Phase 2)
NLP	Tag aggregation, review sentiment (Phase 3)

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12. Out of Scope (Phase 1)

- User authentication
- Real-time retraining
- Dark mode
- Social / sharing features
- Mobile app
- TMDB poster API calls